Open problems:
Physical vector spaces for quantum mechanics

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked [BLOCKS VERIFICATION] stand in the way of checking a main claim of the brief as the brief states it: a *formalization gap* can be closed in Lean alone, while a *brief issue* needs the brief to be corrected or made precise first.

From the brief

1. **Completeness of the Lie algebra.** *Research.* Definition 1 admits pathological cases: a Hilbert space with the defining observables of a qubit ("too many vectors") and a Schwartz space with position as the only defining observable ("too few observables"). "Addressing this gap is an open problem"; the brief expects a "final definition" once it is solved. (Brief, Sec. 5, "Completeness of the Lie algebra"; Sec. 6.)

2. **Isomorphisms of quantum vector spaces.** The brief speaks of isomorphisms "in the category of quantum vector spaces" that "must preserve the Lie algebraic structure", but does not define the morphisms. For a change of frame on the same space, the formalization shows that the identity preserves the topology, the inner product and all observables exactly when the defining observables of each frame are observables in the other. (Brief, Example 3; Theorem 5, proof; Log, "Differences", last two items.)

3. $\mathcal{S}(\mathbb{R}^n) \cong \mathcal{S}(\mathbb{R}^m)$ **iff** $n = m$**.** Stated in Example 3 and argued only by "no linear transformation can change the number of defining observables", although the number of defining observables of a presentation is not shown to be an invariant. Not formalized. (Brief, Example 3 and Remark; Log, "Differences", last item.)

4. **Schwartz spaces as topological vector spaces.** "Schwartz spaces of different dimensionality are isomorphic as topological vector spaces" is stated without proof or reference. (Brief, Remark after Example 3.)

5. **Lie algebras determine physical systems.** *Research.* "Two systems are physically equivalent if and only if their Lie algebras of observables are equivalent" is asserted, and is what makes QSTO:V consistent with [B]STDIST. (Brief, Theorem 5, proof; Log, "Differences", last item.)

6. **Ensembles on a quantum vector space.** *Research.* QSTO:V uses density operators ρ and $S(\rho) = -\operatorname{tr}(\rho \log \rho)$, and Sec. 5 calls the set of ensembles "the most natural space to consider", but no ensemble space over V is defined. In particular, whether the ensembles induce on the pure states the quotient topology of V, under which the pure states are second countable exactly when V is, is open. (Brief, Sec. 3, QSTO:V; Sec. 5.)

7. **Field theories.** *Research.* "For a field theory, we would expect the defining observables to be at least countable"; no field-theory example of a quantum vector space is given. (Brief, Sec. 5, "Number of defining observables".)

8. **Which observables generate unitary groups.** *Research.* Since symmetric and self-adjoint coincide on V, "not all self-adjoint operators generate a unitary transformation", and "to understand which ones the integrability of the system must be checked". A characterization, analogous to complete vector fields in Hamiltonian mechanics, is not given. (Brief, Sec. 5, "Symmetric vs self-adjoint".)

From the formalization

None. Definition 1 and Examples 3 (in n dimensions) and 4 are checked in full.